Dhruv **Menon**

Doctoral Candidate, University of Cambridge · Molecular Design · Generative Deep Learning

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Education

University of Cambridge *Cambridge, United Kingdom*
PhD in Chemical Engineering *2023 - Jan, 2027 (Expected)*

- Areas of Research: generative deep learning, chemical language models, and molecular simulation for inverse design of nanoporous materials.

University of Cambridge *Cambridge, United Kingdom*
M.Res in Physical Sciences (Nanoscience and Nanotechnology) (Grade: 81/100 - Distinction) *2022 - 2023*

- Thesis: Guiding the rational design of biocompatible metal-organic frameworks for drug delivery using interpretable machine learning.

Indian Institute of Technology Gandhinagar *Gandhinagar, India*
B.Tech (with Honours) in Materials Science and Engineering (GPA: 9.11/10.00 - Department Rank 1) *2018 - 2022*

- Institute Gold Medal, Director's Silver Medal, Gold Medal for Outstanding Research (Undergraduate)
- Academic distinctions: Scholarship for Academic Excellence (multiple years, highest GPA); Dean's List (multiple semesters)

Awards

2025	**Winton Cambridge - Berkeley Exchange Award.** Awarded a research fellowship at University of California, Berkeley under the supervision of Professor Omar Yaghi (Nobel Prize in Chemistry, 2025) to work on the "generative design of metal-organic framework (MOF) water harvestors".	*California, US*
2022	**Gold Medal for Outstanding Research (B.Tech).** Awarded for the best overall undergraduate research at IIT Gandhinagar.	*Gandhinagar, India*
2022	**Institute Gold Medal (Materials Science and Engineering).** Awarded for the highest overall GPA in the B.Tech program in Materials Science and Engineering at IIT Gandhinagar.	*Gandhinagar, India*
2022	**Engineering and Physical Sciences Research Council (EPSRC) PhD Award.** Awarded a fully-funded integrated MRes/PhD studentship at the Department of Physics, University of Cambridge.	*Cambridge, UK*
2022	**Trinity-Henry Barlow (Honourary) Scholarship 2022-23.** Awarded on the recommendation of the Trinity-Henry Barlow electors at Trinity College, Cambridge.	*Cambridge, UK*
2020	**Summer Undergraduate Research Fellowship (SURF), Caltech.** Awarded a research fellowship at California Institute of Technology under the guidance of Professor Austin Minnich to work on the "*reduction of electronic noise in semiconductors*." [held virtually due to COVID-19]	*California, US*

Key Research Contributions

Peer-reviewed publications and preprints in Advanced Materials, Chem, Matter, Nature Chemistry, Nature Protocols, Chemical Society Reviews, and others. Full list on Google Scholar. † - equal contribution; ‡ - cover.

1. **Menon, D.**, Singh, V., Chen, X., Alizadeh Kiapi, M.R., Zyuzin, I., Macleod, H.W., Rampal, N., Shepard, W., Yaghi, O.M., and Fairen-Jimenez, D., 2026. **A chemical language model for reticular materials design.** arXiv preprint arXiv:2603.20389 [Link] (JACS, *under review*)

2. **Menon, D.**, Bhadane, P., Mahato, P., Goyal, P., Lynch, I., Chakraborty, S., and Misra, S.K., **Selective heavy-metal sequestration and rare-earth element recovery using metal-organic frameworks.** Nature Protocols (*Accepted in-principle*).

3. Zhao, Q.†, Li, B.†, **Menon, D.**†, Zhu, C.†, Alizadeh Kiapi, M.R., Chen, X., Long, D.-L, Liu, J., Xiao, Z., Yang, D., Fairen-Jimenez, D., Zang, H-Y., and Xuan, W., 2026. **Trigonal-Shaped Cationic Tectons Direct Anion Templated Assembly of Superprotonic Polyanion-Organic Frameworks for High-Power Fuel Cells.** Nature Chemistry, p.1-11.[Link]

4. Melle, F., **Menon, D.**, Conniot, J., Ostolaza-Paraiso, J., Mercado, S., Oliveira, J., Chen, X., Mendes, B.B., Conde, J., and Fairen-Jimenez, D., 2024. **Rational Design of Metal-Organic Frameworks for Pancreatic Cancer Therapy: From Machine Learning Screening to *In Vivo* Efficacy.** Advanced Materials, p.2412757.‡ [Link]

5. **Menon, D.** and Fairen-Jimenez, D., 2025. **Guiding the rational design of biocompatible metal-organic frameworks for drug delivery.** Matter [Link]

6. Chen, X.†, **Menon, D.**†, Wang, X., He, M., Alizadeh Kiapi, M.R., Asgari, M., Lyu, Y., Tang, X., Keenan, L.L., Shepard, W., Wee, L.H., Yang, S., Farha, O.K., and Fairen-Jimenez, D., 2024. **Flexibility-frustrated porosity for enhanced selective CO_2 adsorption in an ultramicroporous metal-organic framework.** Chem‡ [Link]

Selected Other Publications

Selected collaborative publications spanning computational design, mechanistic analysis, and experimental validation of nanoporous materials. This includes work I was involved in as an undergraduate student. Full list on Google Scholar. † - equal contribution.

1. Chakraborty, S., Mikulska, I., Dhumal, P., Langford, N., Nehzati, S., Boseley, R., Pham, S., Pfrang, C., Kaur, M., Valsami-Jones, E., Ignatyev, K., **Menon, D.**, Misra, S.K. and Lynch, I., 2025. **Mapping the Hierarchical Environmental Transformations of Nanoscale UiO-66 Metal–organic Framework.** Environmental Science & Technology. [Link]
2. Kumar, A., Satpathy, B.K., Goyal, P., Upadhyay, R., Alizadeh Kiapi, M.R., Jasuja, K., **Menon, D.** and Misra, S.K., 2025. **Bimetallic MOF-derived CuO-Co_3O_4 heterostructures as high-capacity electrodes for asymmetric supercapacitors.** Chemical Engineering Journal, 520, p.165685. [Link]
3. Zhuang, Y., Mendes, B.B., **Menon, D.**, Oliveira, J., Chen, X., Duman, F.D., Conniot, J., Mercado, S., Liu, X., Zhang, S.Y. and Conde, J., 2025. **Multiscale Profiling of Nanoscale Metal-Organic Framework Biocompatibility and Immune Interactions**. Advanced Healthcare Materials, 14(29), p.e01809.[Link]
4. Bhadane, P.†, **Menon, D.**†, Goyal, P., Alizadeh Kiapi, M.R., Satpathy, B.K., Lanza, A., Mikulska, I., Scatena, R., Michalik, S., Mahato, P., Asgari, M., Chen, X., Chakraborty, S., Mishra, A., Lynch, I., Fairen-Jimenez, D., and Misra, S.K. 2024. **A Two-Dimensional Metal-Organic Framework for Efficient Recovery of Heavy and Light Rare Earth Elements from Electronic Wastes.** Separation and Purification Technology, 360, p.130946. [Link]
5. Goyal, P., Paruthi, A., **Menon, D.**, Behara, R., Jaiswal, A., Keerthy, V., Kumar, A., Krishna, V. and Misra, S.K., 2021. **Fe doped bimetallic HKUST-1 MOF with enhanced water stability for trapping Pb(II) with high adsorption capacity.** Chemical Engineering Journal, p.133088. [Link]

Reviews, Perspectives and Commentaries

Key peer-reviewed non-primary publications. Full list on Google Scholar. ‡ - cover.

1. Asgari, M., Albacete, P., **Menon, D.**, Lyu, Y., Chen, X., and Fairen-Jimenez, D., 2025, **The structuring of porous reticular materials for energy applications at industrial scales.** Chemical Society Reviews.‡ [Link]
2. Chakraborty, S., **Menon, D.**, Mikulska, I., Pfrang, C., Fairen-Jimenez, D., Misra, S.K. and Lynch, I., 2025. **Make metal–organic frameworks safe and sustainable by design for industrial translation.** Nature Reviews Materials, pp.1-3. [Link]
3. **Menon, D.*** and Chakraborty, S., 2023. **How safe are nanoscale metal-organic frameworks?** Frontiers in Toxicology, 5, p.1233854. [Link]
4. **Menon, D.** and Ranganathan, R., 2022. **A Generative Approach to Materials Discovery, Design and Optimization.** ACS Omega, 7(30), pp.25958-25973. [Link]

Seminars

1. ***De novo* design of metal-organic frameworks using generative deep learning**, Kavli ENSI Seminar, UC Berkeley, September 2025.$^{\Phi}$

Φ - seminar/talk

Research Experiences

Research at the intersection of generative machine learning, computational chemistry, and nanoporous materials design. Focused on (i) inverse design of reticular materials using chemical language models and flow-based methods, and (ii) developing mechanistic insights into material performance via molecular simulation. Undergraduate and master's research involved substantial experimental work.

Generative design of metal-organic framework water harvestors — *University of California Berkeley*

Supervisor: Professor Omar Yaghi; Winton - Berkeley Exchange, 2025 — *July - September, 2025*

- Developed **Nexerra**^{R1}, a generative framework combining chemical language models and flow-based methods for inverse design of MOF linkers for water harvesting.
- Established a generative reticular design pipeline integrating representation learning with structure assembly

Generative modeling and computational design in reticular chemistry — *University of Cambridge*

Supervisor: Professor David Fairen-Jimenez; PhD Project — *2023 – Present*

- Develop chemical language models and flow-based generative methods for inverse design of MOF linkers and reticular materials.
- Couple generative modeling with molecular simulation (GCMC, MD) to enable high-throughput screening and uncover structure–property relationships in applications including drug delivery, gas storage, and proton transport.

Defect Engineering of Cu(II) MOFs for Water Remediation — *IIT Gandhinagar*

Supervisor: Professor Superb Misra; Undergraduate Research — *2021-23*

- Developed defect engineering strategies (metal doping, linker exchange) to enhance hydrolytic stability of Cu-based MOFs.
- Demonstrated improved performance for selective capture of toxic heavy metal contaminants from water

Plasmonic Nanostructures for Nanophotonic Devices *University of Cambridge*
Supervisor: Professor Jeremy Baumberg; M.Res Research *2023*

- Designed nanoparticle architectures (dimer formation, ion-beam milling) to enhance plasmonic coupling in nano-antennae.
- Investigated structure–property relationships governing optical response in nanophotonic systems.

Biohybrid Systems for Artificial Photosynthesis *University of Cambridge*
Supervisor: Professor Erwin Reisner; M.Res Research *2023*

- Developed biohybrid electrode systems using adaptive laboratory evolution of syngas-fermenting bacteria.
- Enabled improved bio-photoelectrochemical performance through enhanced catalyst–electrode interfacing.

Generative Approaches to Materials Discovery, Design and Optimisation *IIT Gandhinagar*
Supervisor: Professor Raghavan Ranganathan; Undergraduate Research *2021-22*

- Developed early generative modeling approaches (VAEs, GANs) for materials design.
- Explored representation learning strategies for navigating materials design spaces.

Peer Review

Reviewer for leading journals including Chemical Society Reviews, Advanced Science, Journal of Chemical Theory and Computation, Materials Horizons, and Nature Communications (co-reviewer).

Supervision & Teaching

Undergraduate Supervisor *University of Cambridge*
Department of Chemical Engineering and Biotechnology *2025-2026*

- Adsorption and Advanced Nanoporous Materials (final-year, IIB)
- Materials Science (first-year, IA)

Co-supervision *University of Cambridge*
Department of Chemical Engineering and Biotechnology *2023-2024*

- Supervised MPhil research project on machine learning for CO_2 sorbents.
- Supervised undergraduate research project on machine learning for MOF ecotoxicology

Teaching Assistant *IIT Gandhinagar*
Department of Materials Science and Engineering *2021*

- Delivered instructional sessions and developed course materials for $>$170 students in ES202 - Introduction to Materials

Technical Knowledge

Machine Learning: generative modeling (VAEs, transformers, graph neural networks, flow matching), representation learning
Programming: Python, PyTorch, RDKit, PyTorch-Geometric (intermediate)
Computational Chemistry: GCMC, molecular dynamics, adsorption modeling
Simulation Tools: RASPA, LAMMPS, Materials Studio

Additional

Football Player *IIT Gandhinagar*
IIT Gandhinagar Football Team *2018-22*

- Represented the institute at the 54th Inter-IIT Sports Meet at IIT Kharagpur in 2019 in Men's Football and various district- and state-level college tournaments.

References

Professor David Fairen-Jimenez *University of Cambridge*
Professor of Molecular Engineering *Cambridge, United Kingdom*

Professor Omar M. Yaghi *University of California, Berkeley*
The James and Neeltje Tretter Professor of Chemistry *Berkeley, United States*

Professor Iseult Lynch *University of Birmingham*
Professor of Environmental Nanosciences *Birmingham, United Kingdom*

Professor Superb K. Misra *IIT Gandhinagar*
Jibaben Patel Chair Professor *Gandhinagar, India*

Dr Swaroop Chakraborty *University of Birmingham*
NERC Independent Research Fellow *Birmingham, United Kingdom*

Professor Weimin Xuan *Donghua University*
Professor of Chemistry and Chemical Engineering *Shanghai, China*